Quantum Network and Repeater Simulation Tutorial

Agenda

Session 1 (*90 min*):

- ***Tutorial session introduction (10m) - Ezra Kissel***
  - Overview of agenda and objectives
  - Introduction to tutorial exercises JH environment

- ***Introduction to NetSquid (45 min) - Michal van Hooft***
  - Setup and configuration
  - Code demonstrations
  - Simulation of a simple quantum network

- ***Exercise 1: Setup and configuration (15 min) - Michal van Hooft***
  - Attendees will learn to setup and use a NetSquid installation
  - Attendees will run a simple simulation of quantum network

- ***Tools built on top of NetSquid (10 min) - Michal van Hooft***
  - Installing Netsquid
  - Netsquid documentation
  - Tools built on top of NetSquid, Overview of: NetSquid-snippets, SquidASM and NetSquid-netbuilder

- ***Questions (5 min)***

Session 2 (*90 min*):

- ***Modeling and simulation of "all-photonic" quantum repeaters and networks (30 min) - Cheun-Hei Chan***
  - Attendees will be introduced to the quantum repeater simulation toolkit
- ***Exercise 2: simulation of all-photonic quantum repeaters and networks (10 min) - Cheun-Hei Chan***
  - Attendees will run a simple simulation of all-photonic quantum network
- ***Modeling and simulation of "memory-based" trapped ion quantum repeaters and networks (20 min) - Charu Jain***
- ***Exercise 3: simulation of trapped ion quantum repeaters and networks (10 min) - Charu Jain***
  - Attendees will run a simple simulation of trapped-ion quantum network
- ***Exploring quantum data center models and distributed quantum computing in simulation (15 min) - Charu Jain***
  - *Point to DQC exercise/examples*
- ***Wrap up (5 min)***